FLASH DROUGHT PROJECT MEETING #10

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- Jess: Naomi (fire team) is doing some case studies from a different perspective. Jess has run some similar case studies for comparison.
- Jess gave an AMOS presentation on the flash drought work. Intends to do a draft of the reanalysis results before she goes on long leave in June.
- Jess presented updates – and has sent us the slides
- Driver sensitivities for bi-annual seasons – Mike: how were these aggregated? Jess: Averaged daily sensitivities. Will check the method in Mike's paper – alternative is to aggregate inputs.
- Jess: relative driver contribution plots – shows dominating driver seasonally.
- Mike: suggest limiting the domain to the Australian continent, will reveal more detail in Aus as some results in the Maritime Continent are skewing the contour levels.
- Jess: Tinderbox Drought decomposition – showed results spatially, by driver rank, and temporally.
- 2025 Jan – May flash drought (anecdotal evidence of stressed crops). Was the warmest autumn on record.
- Black Saturday bush fire example: Kinglake vs Beechworth decompositions – wind was more of a factor in the latter, but geography/terrain may play a role here.
- David notes that BARRA-R2 wind is not the best, maybe try BARRA-C2.
- Role of wind in fire is more around fire spread than contribution to drying. BASS (?) fire dataset – Mike will look at the wind signal in these fires.
- Tess: try to model the event with and without wind? Mike: reanalysis wind is terrible. Take the observed winds over period of fire and use those winds instead perhaps.
- Convective-permitting or km scale model may be useful.
- Mike: question of how we use the results? We've deconstructed the demand side of the drought – so what? Predictability and predictive skill of each of these drivers: if wind isn't predictable, and we don't have predictive skill in wind, that doesn't help. We have the machinery of what drives drought prior to fires, and the contribution from each driver. Now if we can predict each of those drivers, forecasts of opportunity for each of these drivers, this is the operational attraction. How do we make this usable?
- Kangaroo Island example: closure area larger, may be due to maritime pixels.
- Subtropical regions are of interest to the fire team – Black Summer, Daintree Forest fires – maybe also NT grass fires?
- WA fire example: contribution from wind more dominant.
- Tess: maybe look at the progression of synoptic systems and soil moisture, as was done for the Wimmera FD, for insight into the changing role of various drivers over a timeline of several months.

- Jess: projections - drivers are available downscaled and bias adjusted. Interested in looking at analogous events in projections in hotter, more variable ET models.
- Mike: could decompose the variability of ET and look at which of the drivers is responsible for that. But may be a lot of work.
- Tess: ECMWF have updated their TIGGE archive of NWP ensembles- useful for UNSEEN experiments, etc. Much easier to access these data now.
- Questions from AMOS (Ailie): process versus response? Maybe covariances would be helpful?
- CO_2 included ET equation? Mike will ask a colleague.
- David used a paper for aridity work - https://iopscience.iop.org/article/10.1088/1748-9326/ab5046 Would require redoing calculus.